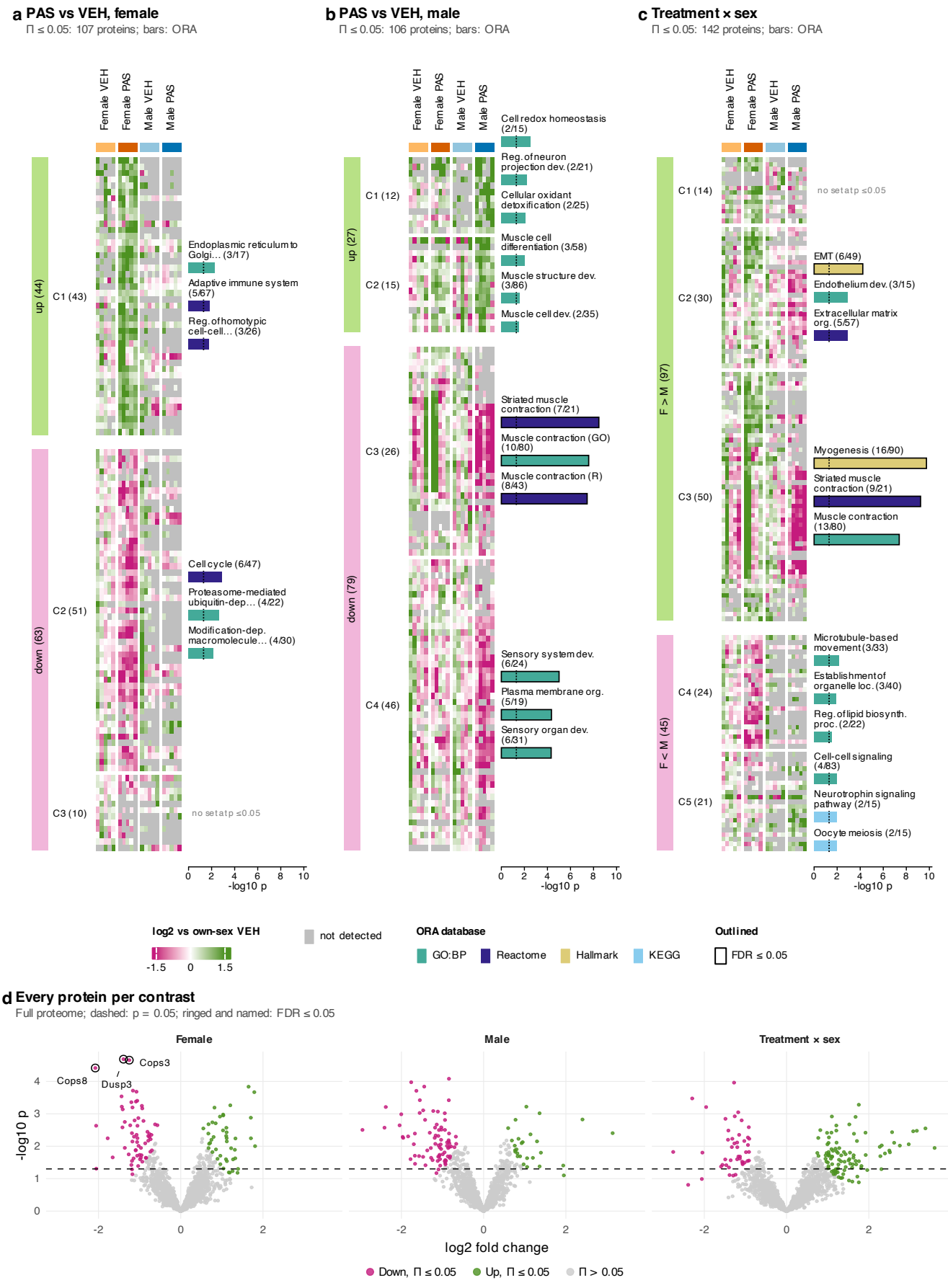

a PAS vs VEH, female
Π ≤ 0.05: 107 proteins; bars: ORA
Female VEH
Female PAS
Male VEH
Male PAS
up (44)
C1 (43)
Endoplasmic reticulum to Golgi… (3/17)
Adaptive immune system (5/67)
Reg. of homotypic cell-cell… (3/26)
down (63)
C2 (51)
Cell cycle (6/47)
Proteasome-mediated ubiquitin-dep… (4/22)
Modification-dep. macromolecule… (4/30)
C3 (10)
no set at p ≤0.05
0 2 4 6 8 10
-log10 p
b PAS vs VEH, male
Π ≤ 0.05: 106 proteins; bars: ORA
Female VEH
Female PAS
Male VEH
Male PAS
up (27)
C1 (12)
Cell redox homeostasis (2/15)
Reg. of neuron projection dev. (2/21)
Cellular oxidant detoxification (2/25)
C2 (15)
Muscle cell differentiation (3/58)
Muscle structure dev. (3/86)
Muscle cell dev. (2/35)
down (79)
C3 (26)
Striated muscle contraction (7/21)
Muscle contraction (GO) (10/80)
Muscle contraction (R) (8/43)
C4 (46)
Sensory system dev. (6/24)
Plasma membrane org. (5/19)
Sensory organ dev. (6/31)
0 2 4 6 8 10
-log10 p
c Treatment × sex
Π ≤ 0.05: 142 proteins; bars: ORA
Female VEH
Female PAS
Male VEH
Male PAS
F > M (97)
C1 (14)
no set at p ≤0.05
C2 (30)
EMT (6/49)
Endothelium dev. (3/15)
Extracellular matrix org. (5/57)
C3 (50)
Myogenesis (16/90)
Striated muscle contraction (9/21)
Muscle contraction (13/80)
F < M (45)
C4 (24)
Microtubule-based movement (3/33)
Establishment of organelle loc. (3/40)
Reg. of lipid biosynth. proc. (2/22)
Cell-cell signaling (4/83)
C5 (21)
Neurotrophin signaling pathway (2/15)
Oocyte meiosis (2/15)
0 2 4 6 8 10
-log10 p
log2 vs own-sex VEH
-1.5 0 1.5
not detected
ORA database
GO:BP
Reactome
Hallmark
KEGG
Outlined
FDR ≤ 0.05
d Every protein per contrast
Full proteome; dashed: p = 0.05; ringed and named: FDR ≤ 0.05
Female
Male
Treatment × sex
Cops3
Cops8
Dusp3
-log10 p
log2 fold change
Down, Π ≤ 0.05
Up, Π ≤ 0.05
Π > 0.05